If you are wondering why we can rely on these distances and how we can come up with a good value for T_2 (and T_1), you are on top of the main issue: these parameters usually need to be tuned (trying a few different ones and checking the number/quality of the canopies), and sometimes good initial estimates for these distances come from experience and domain knowledge; for instance, if you were clustering geographical data about cellphone cells and you knew that no two cells are further away than a few kilometers, you would have a hint about T_1.

Anyway, deciding these values is not such a big deal as it could seem, it is not important to come up with the best possible value, but rather to find an acceptable one for both parameters: as we mentioned, this algorithm only provides a coarse-grain clustering.

13.1.4 Applying Canopy Clustering

Canopy clustering, in fact, is often used as a pre-processing step for k-means, but it can also be used for DBSCAN and OPTICS: as a matter of facts, for those algorithms it has one further advantage. But we'll get there in a moment: first, let's talk about how to combine canopy clustering and k-means.

The easiest way is quite intuitive: we can take the coarse-grain clusters (with overlap) outputted by canopy clustering, and for each of them compute their center of mass. Since these clusters can overlap each other, some points will belong to more than one of them: we can't, thus, just treat these canopies as the result of an iteration of k-means!